

CHE 201

1. Course number and name

CHE 201. Introduction to Thermodynamics

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. Instructor's or course coordinator's name

Betul Bilgin, Alan Zdunek

4. Text book, title, author, publisher, year

Fundamentals of Engineering Thermodynamics, Moran/Shapiro/Boettner/Bailey, 9th Edition (2018)

5. Specific course information

a. Brief description of the content of the course (catalog description)

Work and energy; conversion of energy; theory of gases and other states of matter; applications to energy conversion devices. Second law of thermodynamics, entropy, and equilibrium, with applications.

b. Prerequisites or co-requisites

Math 181, Calculus II and Phys 141, General Physics I (Mechanics).

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction.

Students will learn to:

- Apply energy balances for closed systems and control volumes.
- Evaluate thermodynamic efficiencies of compressors, turbines, power plants, refrigeration and heat pump cycles.
- Identify phases at a given state and find thermodynamic properties such as internal energy, specific volumes from steam tables.
- Draw and interpret pressure-enthalpy (P-H), pressure-specific volume, temperature specific volume, temperature-entropy charts.
- Relate thermodynamic properties via partial derivative, Maxwell's relations.
- Interpret phase diagrams of binary systems.
- Apply the entropy balance and identify if the process is reversible or irreversible.

Project

One, semester long project related to thermodynamics concept. Students will be working with groups and are expected to design a calorimeter. Project includes research phase, modeling phase, prototype phase and testing phase. Design competition will be held among all the groups. All the groups are required to create Adobe Spark webpage for their design.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome 1 Not Covered, 2 Introduced, 3 Reinforced, 4 Emphasized	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.			X	
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.		X		
(3) an ability to communicate effectively with a range of audiences.		X		
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.		X		
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.		X		
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.		X		

7. Brief list of topics to be covered

- I. Introduction
- II. Fundamentals (Chapter 1)
- III. The First Law (Chapter 2)
- IV. Properties of Pure Fluids (Chapter 3)
- V. Control Volume Analysis (Chapter 4)
- VI. The Second Law of Thermodynamics (Chapter 5)
- VII. Entropy (Chapter 6)
- VIII. Thermodynamic Relations (Chapter 11 Section 1 through 5)
- IX. Power Cycles (Chapter 9; select sections)

SaChE safety modules:

Students are expected to complete SaChE safety modules online and submit their certificates of completion to the instructor. There are only two relevant SaChE modules for CHE 201:

- The importance of Process Safety
- Hazard Recognition